

CUSTOMER BEHAVIOUR ANALYSIS REPORT

A comprehensive data analytics project exploring shopping patterns, customer segmentation, and revenue insights

3,900

Total Customers

3.75 / 5

Avg Review Rating

\$59.76

Avg Purchase Amount

1. Project Overview

This report presents a comprehensive analysis of customer shopping behaviour using a dataset of 3,900 customers. The analysis covers purchase patterns, demographic insights, subscription behaviour, and product performance across multiple categories. The goal is to extract actionable insights to support business decisions in marketing, inventory, and customer retention.

Dataset Summary

Attribute	Details
Total Records	3,900 customers
Age Range	18 – 70 years
Purchase Amount Range	\$20 – \$100
Review Rating Range	2.5 – 5.0
Previous Purchases Range	1 – 50
Categories	Clothing, Accessories, Footwear, Outerwear
Shipping Types	2-Day, Express, Free, Next Day Air, Standard, Store Pickup
Key Fields	Gender, Age, Item Purchased, Category, Location, Season, Subscription Status, Discount Applied, Promo Code Used

2. Analytical Questions & Findings

The following 10 analytical questions were investigated using SQL queries on the customer dataset.

Q1: Total Revenue by Gender

What is the total revenue generated by male vs. female customers?

Gender	Total Revenue (USD)
Male	\$157,890
Female	\$75,191

Key Finding

Male customers contributed significantly more revenue (\$157,890) compared to female customers (\$75,191) — roughly a 2:1 ratio. This may reflect a higher proportion of male customers in the dataset or higher average spend per transaction.

Q2: Customers Using Discounts Who Spent Above Average

Which customers used a discount but still spent more than the average purchase amount (\$59.76)?

A sample of customers who applied a discount code yet still exceeded the \$59.76 average purchase amount:

Customer ID	Purchase Amount (USD)
2	\$64
3	\$73
4	\$90
7	\$85
9	\$97
12	\$68
13	\$72
16	\$81

Key Finding

A notable segment of customers still spend above average even when discounts are applied, indicating strong purchase intent and willingness to pay. These customers represent high-value targets for loyalty programmes.

Q3: Top 5 Products by Average Review Rating

Which are the top 5 products with the highest average review rating?

Rank	Category	Avg Review Rating
1	Footwear	3.79
2	Accessories	3.77
3	Outerwear	3.75
4	Clothing	3.72

Key Finding

Footwear leads in customer satisfaction with an average rating of 3.79, followed closely by Accessories (3.77). All categories maintain ratings between 3.72 and 3.79, indicating consistently positive customer experiences across the board.

Q4: Average Purchase Amount — Standard vs. Express Shipping

Compare the average purchase amounts between Standard and Express shipping types.

Shipping Type	Avg Purchase Amount (USD)
Standard	\$58.46
Express	\$60.48

Key Finding

Customers who choose Express shipping tend to spend slightly more (\$60.48) compared to those opting for Standard shipping (\$58.46). This suggests Express shipping customers may have higher purchasing power or urgency, making them a valuable segment.

Q5: Subscriber vs. Non-Subscriber Spending

Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

Subscription Status	Avg Purchase Amount (USD)
No (Non-Subscriber)	\$59.87
Yes (Subscriber)	\$59.49

Key Finding

Interestingly, non-subscribers (\$59.87) spend marginally more per transaction than subscribers (\$59.49). The difference is minimal (~\$0.38), suggesting that subscription status alone does not drive higher individual transaction values. However, subscribers likely generate more lifetime value through repeat purchases.

Q6: Top 5 Products with Highest Discount Rate

Which 5 products have the highest percentage of purchases with discounts applied?

Rank	Product	Discount Rate (%)
1	Hat	50.00%
2	Sneakers	49.00%
3	Coat	49.00%
4	Sweater	48.00%
5	Pants	47.00%

Key Finding

Hats have the highest discount application rate at 50%, meaning half of all Hat purchases included a discount. This high rate across these products may indicate they are used as promotional drivers or that customers in these categories are highly price-sensitive.

Q7: Customer Segmentation — New, Returning & Loyal

Segment customers into New (1–5 previous purchases), Returning (6–20), and Loyal (>20) based on their total number of previous purchases.

Segment	Definition	Customer Count
New	1–5 previous purchases	To be queried
Returning	6–20 previous purchases	To be queried
Loyal	More than 20 previous purchases	To be queried

Key Finding

The dataset's statistics show that the median customer has 25 previous purchases (50th percentile), with 75% of customers having up to 38 purchases. This suggests a majority of customers fall in the 'Loyal' category, indicating a strong repeat-purchase customer base. The average is 25.35 previous purchases per customer.

Q8: Top 3 Most Purchased Products per Category

What are the top 3 most purchased products within each category?

Sample from Accessories category:

Rank	Product	Purchase Count
1	Blouse	171
2	Belt	161
3	Backpack	143

Key Finding

Within the Accessories category, Blouses (171), Belts (161), and Backpacks (143) are the most purchased items. This data helps identify which SKUs drive category volume and should be prioritised for stock management and promotional campaigns.

Q9: Repeat Buyers and Subscription Likelihood

Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

Methodology

Customers with more than 5 previous purchases were analysed against their subscription status (Yes/No) to determine if repeat purchase behaviour correlates with subscription uptake.

Key Finding

The dashboard shows that overall, 27% of all customers are subscribers and 73% are non-subscribers. Given that the average customer has 25+ previous purchases, it suggests that even loyal repeat buyers do not automatically convert to subscribers. This highlights an opportunity to improve subscription conversion among the high-repeat-purchase segment through targeted campaigns.

Q10: Revenue Contribution by Age Group

What is the revenue contribution of each age group?

Rank	Age Group	Total Revenue (USD)
1	Young Adult	\$62,143
2	Middle Age	\$59,197
3	Adult	\$55,978
4	Senior	\$55,763

Key Finding

Young Adults are the highest revenue-generating age group at \$62,143, followed by Middle Age (\$59,197), Adults (\$55,978), and Seniors (\$55,763). The revenue is relatively evenly distributed across age groups, but Young Adults represent the most lucrative segment for targeted marketing spend.

3. Power BI Dashboard Overview

The interactive Power BI dashboard titled 'Customer Behaviour' provides a visual summary of the dataset with the following key components:

Visual	Description
KPI Cards	Displays 3 key metrics: 4K customers, 3.75 avg rating, \$59.76 avg purchase
Total % of Subscription	Donut chart — 73% No, 27% Yes subscriptions
Revenue of Each Product	Bar chart by category: Clothing \$104K, Accessories \$74K, Footwear \$36K, Outerwear \$19K
Sales by Category	Count of sales: Clothing 1,737 Accessories 1,240 Footwear 599 Outerwear 324
Revenue by Age Group	Horizontal bars: Young Adult \$62K > Middle Age \$59K > Adult \$56K > Senior \$56K
Sales by Age Group	Horizontal bars: Young Adult 1,028 > Middle Age 986 > Senior 944 > Adult 942
Filter Panel	Slicers for Subscription Status, Gender, Category, and Shipping Type



4. Key Insights & Recommendations

#	Insight	Recommendation
1	Clothing is the top revenue category (\$104K)	Invest more in Clothing inventory and promotions
2	Young Adults drive the most revenue (\$62K)	Target Young Adult segment with digital marketing campaigns
3	Only 27% of customers subscribe	Launch targeted subscription conversion campaigns for repeat buyers
4	Express shipping users spend ~\$2 more per order	Promote Express shipping with bundle offers to increase AOV
5	50% of Hat purchases use discounts	Evaluate discount sustainability for high-discount products
6	Male customers generate 2x more revenue than female	Investigate female customer engagement and personalise offers
7	Footwear has the highest review rating (3.79)	Use Footwear as a brand ambassador product in marketing
8	Revenue is evenly spread across all age groups	Develop age-specific product lines to capture niche segments

5. Statistical Summary

Descriptive statistics for key numerical fields across all 3,900 customer records:

Metric	Customer ID	Age	Purchase Amt	Review Rating	Prev. Purchases	Freq. Days
Count	3,900	3,900	3,900	3,900	3,900	3,900
Mean	1,950.5	44.07	\$59.76	3.75	25.35	89.13
Std Dev	1,125.98	15.21	\$23.69	0.71	14.45	119.04
Min	1	18	\$20	2.5	1	7
25th %ile	975.75	31	\$39	3.1	13	14
Median	1,950.5	44	\$60	3.8	25	30
75th %ile	2,925.25	57	\$81	4.4	38	90
Max	3,900	70	\$100	5.0	50	365

6. Conclusion

This Customer Behaviour Analysis project successfully explored a dataset of 3,900 customers using SQL-based queries and Power BI visualisations. The analysis revealed critical insights about purchasing patterns, demographic contributions, product performance, and subscription behaviour.

Key takeaways include:

- Clothing dominates both revenue and sales volume across all categories.
- Young Adults are the most valuable demographic segment by revenue contribution.
- Subscription rates remain low at 27%, even among repeat buyers — presenting a growth opportunity.
- Gender disparity in revenue (Male 2x Female) warrants targeted female customer engagement strategies.
- Discount strategies, particularly for Hats and Sneakers, should be reviewed for profitability.
- Customer satisfaction is consistently high across all categories (3.72–3.79 rating).

These findings provide a solid foundation for data-driven decision-making in marketing, inventory management, customer retention, and product strategy.

Customer Behaviour Analysis

Data Analytics Project | 3,900 Customer Records | Power BI + SQL